



10893 - A High-Precision Test of Substellar Cooling Models with the Benchmark System HR 7672AB

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
NIRCam Coronagraphy				
	1	HR7672-Roll1	NIRCam Coronagraphic Imaging	(1) HR7672
	2	HR7672-Roll2	NIRCam Coronagraphic Imaging	(1) HR7672
	3	HR7672-PSF-REF	NIRCam Coronagraphic Imaging	(2) HR7672-PSF-REFERENCE
NIRCam Direct Imaging				
	4	HR7672-F162M-Roll1	NIRCam Time Series	(1) HR7672
	5	HR7672-F140M-Roll1	NIRCam Time Series	(1) HR7672
	6	HR7672-F115W-Roll1	NIRCam Time Series	(1) HR7672
	7	HR7672-F162M-Roll2	NIRCam Time Series	(1) HR7672
	8	HR7672-F140M-Roll2	NIRCam Time Series	(1) HR7672
	9	HR7672-F115W-Roll2	NIRCam Time Series	(1) HR7672

ABSTRACT

Substellar evolutionary models describe how brown dwarfs and gas giant planets cool and fade as a function of mass and age, and thus are fundamental for converting observations into the physical quantities needed for science. Growing evidence points to a "cooling-age problem," namely that the ages inferred for substellar companions from their dynamical masses and bolometric luminosities are in discord with ages independently determined for their host stars, indicating systematic errors in these widely used models. Progress has been hampered primarily by large uncertainties in the host-star ages. We will capitalize on a new high-precision (17%) asteroseismic age for the solar analog HR 7672A and a new exceptionally precise (0.9%) dynamical mass for its L-dwarf companion HR 7672B. We will obtain multi-band (1-5 micron) JWST/NIRCam photometry to reduce the uncertainty in HR 7672B's bolometric luminosity by an order of magnitude (from ~25% to ~2.5%). The resulting unmatched combination of mass, age, and luminosity for this system will enable a unique, high-precision test of substellar evolution, sufficient to discriminate among current model grids at the stellar/substellar boundary and establishing an empirical anchor for present and future modeling of low-mass stars, brown dwarfs, and gas-giant planets.

OBSERVING DESCRIPTION

We will observe for HR 7672B using both direct (1.0-1.7 micron) and coronagraphic imaging (1.8-5.0 micron). Direct imaging is needed at the shortest wavelengths (1.0-1.7 micron), since coronagraphy is not feasible, and we will obtain such data using rapid-readout time-series mode.

For coronagraphy (1.7-5.0 micron), we select the intermediate-sized MASK335R mask to simultaneously detect the companion at high throughput and high contrast in both the short and long wavelength channels of NIRCam, thereby expediting our observing while maximizing our wavelength coverage.

Our program is designed to measure the 1-5 micron SED of this 1.0" companion to the G0V star HR 7672A with a continuous set of medium-band filters (F140M to F480M), with F115W being the only wide-band filter (to cover the shortest wavelengths). Our goal is to achieve 1% photometric precision in each filter (not considering the overall 1% absolute photometric calibration error of NIRCam).

We use standard high-contrast imaging methods for the observing sequences, namely angular and reference differential imaging (ADI and RDI), so our observations include a PSF star. For the PSF reference star, we chose HD 187923, a nearby star of similar spectral type and magnitude to our target. The star has a low Gaia RUWE = 0.8 indicating it is likely a single star, and has no companion in the Washington Double Star catalog.

We require position angle (PA) constraints to place the companion off of the bright host star diffraction spikes in the direct imaging sequences to

JWST Proposal 10893 (Created: Friday, March 13, 2026, 3:02:15PM Eastern Standard Time) - Overview

maximize companion signal-to-noise. For each sequence (direct imaging and coronagraphy), we require relative position angle offsets ranging anywhere from 7-14 degs to perform Angular Differential Imaging (ADI), following best practice recommendations for high contrast science from commissioning and ERS program 1386. We validated our PA constraints using STPSF models and the JWST Coronagraphic Visibility Tool. The observations for each filter must be executed as uninterruptible groups to minimize the wavefront drifts between images and facilitate differential imaging.

Note that the system has ~20 years of AO imaging and ~40~years of RVs, so the orbit is well-known, with a predicted separation range of 0.99-1.02 arcsec (0.03 arcsec uncertainty) and PA range from 329.5 -328.9 deg (0.7 deg uncertainty) during Cycle 5.

Proposal 10893 - Targets - A High-Precision Test of Substellar Cooling Models with the Benchmark System HR 7672AB

	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
Fixed Targets	(1)	HR7672	RA: 20 04 6.2208 (301.0259200d) Dec: +17 04 12.68 (17.07019d) Equinox: J2000	Proper Motion RA: -387.472 mas/yr Proper Motion Dec: -419.4970000298781 mas/yr Parallax: 0.0562693" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Brown dwarfs, G giants]				
Fixed Targets	(2)	HR7672-PSF-REFERENCE	RA: 19 52 3.4384 (298.0143267d) Dec: +11 37 41.97 (11.62833d) Equinox: J2000	Proper Motion RA: -344.042 mas/yr Proper Motion Dec: -334.59999990554934 mas/yr Parallax: 0.0370027" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. This star, the solar analog HR 7569, was selected as a PSF reference because of its on-sky proximity to HR 7672, its similar spectral type and brightness, and low Gaia RUWE (0.9) indicating it is likely not a binary. Category=Star Description=[G dwarfs]				

Proposal 10893 - Observation 1 - A High-Precision Test of Substellar Cooling Models with the Benchmark System HR 7672AB

Observation	Proposal 10893, Observation 1: HR7672-Roll1										Fri Mar 13 20:02:15 GMT 2026	
	Diagnostic Status: Warning											
	Observing Template: NIRCam Coronagraphic Imaging											
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(HR7672-Roll1 (Obs 1)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(1)	HR7672	RA: 20 04 6.2208 (301.0259200d) Dec: +17 04 12.68 (17.07019d) Equinox: J2000			Proper Motion RA: -387.472 mas/yr Proper Motion Dec: -419.4970000298781 mas/yr Parallax: 0.0562693" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Star Description=[Brown dwarfs, G qiants]											
Acquisition	#	Target	Filter	Target Brightness	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID		
	1	SAME	F335M	BRIGHT (ND Square)	SHALLOW4	65	1	1	16.305	271719		
Template	Module		Occulting Mask		Obtain Astrometric Confirmation Images?		Subarray		Dither Pattern			
	A		MASK335R		true		SUB320A335R		NONE			
Confirmation	#	Conf. Readout Pattern		Conf. Groups/Int		Conf. Integrations/Exp		Conf. Total Integrations		Conf. Total Exposure Time		Conf. Total Dithers
	1	RAPID		4		1		1		42.947		1
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID		
	1	F182M	F250M	RAPID	5	4	1	4	25.739			
	2	F182M	F300M	RAPID	5	4	1	4	25.739			
	3	F182M	F335M	RAPID	6	6	1	6	45.023			
	4	F182M	F360M	RAPID	6	4	1	4	30.015			
	5	F182M	F410M	RAPID	8	5	1	5	48.209			
	6	F210M	F430M	RAPID	9	6	1	6	64.265			
	7	F210M	F460M	RAPID	9	6	1	6	64.265			
	8	F210M	F480M	RAPID	10	6	1	6	70.68			

Proposal 10893 - Observation 1 - A High-Precision Test of Substellar Cooling Models with the Benchmark System HR 7672AB

PSF References	HR7672-PSF-REF (Obs 3) (PSF Reference; Filters [F182M/F250M, F182M/F300M, F182M/F335M, F182M/F360M, F182M/F410M, F210M/F460M, F210M/F480M]) HR7672-Roll2 (Obs 2) (Filters [F182M/F250M, F182M/F300M, F182M/F335M, F182M/F360M, F182M/F410M, F210M/F430M, F210M/F460M, F210M/F480M]) Additional Justification: false
Special Requirements	No Parallel Attachments Group Observations 1, 2, 3, Non-interruptible Aperture PA Offset 1 from 2 by 7 to 14 Degrees (Same offsets in V3)

Proposal 10893 - Observation 2 - A High-Precision Test of Substellar Cooling Models with the Benchmark System HR 7672AB

Observation	Proposal 10893, Observation 2: HR7672-Roll2										Fri Mar 13 20:02:15 GMT 2026	
	Diagnostic Status: Warning											
	Observing Template: NIRCam Coronagraphic Imaging											
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(HR7672-Roll2 (Obs 2)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(1)	HR7672	RA: 20 04 6.2208 (301.0259200d) Dec: +17 04 12.68 (17.07019d) Equinox: J2000			Proper Motion RA: -387.472 mas/yr Proper Motion Dec: -419.4970000298781 mas/yr Parallax: 0.0562693" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Star Description=[Brown dwarfs, G qiants]											
Acquisition	#	Target	Filter	Target Brightness	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID		
	1	SAME	F335M	BRIGHT (ND Square)	SHALLOW4	65	1	1	16.305	271719		
Template	Module		Occulting Mask		Obtain Astrometric Confirmation Images?		Subarray		Dither Pattern			
	A		MASK335R		true		SUB320A335R		NONE			
Confirmation	#	Conf. Readout Pattern		Conf. Groups/Int		Conf. Integrations/Exp		Conf. Total Integrations		Conf. Total Exposure Time		Conf. Total Dithers
	1	RAPID		4		1		1		42.947		1
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID		
	1	F182M	F250M	RAPID	5	4	1	4	25.739			
	2	F182M	F300M	RAPID	5	4	1	4	25.739			
	3	F182M	F335M	RAPID	6	6	1	6	45.023			
	4	F182M	F360M	RAPID	6	4	1	4	30.015			
	5	F182M	F410M	RAPID	8	5	1	5	48.209			
	6	F210M	F430M	RAPID	9	6	1	6	64.265			
	7	F210M	F460M	RAPID	9	6	1	6	64.265			
	8	F210M	F480M	RAPID	10	6	1	6	70.68			

Proposal 10893 - Observation 2 - A High-Precision Test of Substellar Cooling Models with the Benchmark System HR 7672AB

PSF References	HR7672-PSF-REF (Obs 3) (PSF Reference; Filters [F182M/F250M, F182M/F300M, F182M/F335M, F182M/F360M, F182M/F410M, F210M/F460M, F210M/F480M]) HR7672-Roll1 (Obs 1) (Filters [F182M/F250M, F182M/F300M, F182M/F335M, F182M/F360M, F182M/F410M, F210M/F430M, F210M/F460M, F210M/F480M]) Additional Justification: false
Special Requirements	No Parallel Attachments Group Observations 1, 2, 3, Non-interruptible Aperture PA Offset 1 from 2 by 7 to 14 Degrees (Same offsets in V3)

Proposal 10893 - Observation 3 - A High-Precision Test of Substellar Cooling Models with the Benchmark System HR 7672AB

Observation	Proposal 10893, Observation 3: HR7672-PSF-REF										Fri Mar 13 20:02:15 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCam Coronagraphic Imaging										
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(2)	HR7672-PSF-REFERENCE	RA: 19 52 3.4384 (298.0143267d) Dec: +11 37 41.97 (11.62833d) Equinox: J2000			Proper Motion RA: -344.042 mas/yr Proper Motion Dec: -334.59999990554934 mas/yr Parallax: 0.0370027" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.										
	This star, the solar analog HR 7569, was selected as a PSF reference because of its on-sky proximity to HR 7672, its similar spectral type and brightness, and low Gaia RUWE (0.9) indicating it is likely not a binary. Category=Star										
	Description=[G dwarfs]										
Acquisition	#	Target	Filter	Target Brightness	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	SAME	F335M	BRIGHT (ND Square)	SHALLOW4	65	1	1	16.305	271719	
Template	Module		Occulting Mask		Obtain Astrometric Confirmation Images?		Subarray		Dither Pattern		
	A		MASK335R		true		SUB320A335R		9-POINT-CIRCLE		
Confirmation	#	Conf. Readout Pattern		Conf. Groups/Int	Conf. Integrations/Exp		Conf. Total Integrations		Conf. Total Exposure Time	Conf. Total Dithers	
	1	RAPID		4	1		1		42.947	1	

Proposal 10893 - Observation 3 - A High-Precision Test of Substellar Cooling Models with the Benchmark System HR 7672AB

Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F182M	F250M	RAPID	5	1	9	9	57.912	
	2	F182M	F300M	RAPID	5	1	9	9	57.912	
	3	F182M	F335M	RAPID	6	1	9	9	67.534	
	4	F182M	F360M	RAPID	6	1	9	9	67.534	
	5	F182M	F410M	RAPID	8	1	9	9	86.777	
	6	F210M	F460M	RAPID	9	1	9	9	96.398	
	7	F210M	F460M	RAPID	9	1	9	9	96.398	
	8	F210M	F480M	RAPID	10	1	9	9	106.019	
PSF References	PSF Reference: true									
Special Requirements	No Parallel Attachments Group Observations 1, 2, 3, Non-interruptible									

Proposal 10893 - Observation 4 - A High-Precision Test of Substellar Cooling Models with the Benchmark System HR 7672AB

Observation	Proposal 10893, Observation 4: HR7672-F162M-Roll1											Fri Mar 13 20:02:15 GMT 2026
	Diagnostic Status: Warning Observing Template: NIRCam Time Series											
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. (HR7672-F162M-Roll1 (Obs 4)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	HR7672	RA: 20 04 6.2208 (301.0259200d) Dec: +17 04 12.68 (17.07019d) Equinox: J2000				Proper Motion RA: -387.472 mas/yr Proper Motion Dec: -419.4970000298781 mas/yr Parallax: 0.0562693" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Star Description=[Brown dwarfs, G qiants]											
Acquisition	#	Target	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID		
	1	SAME	SUB32TATS	F405N+F444W	RAPID	5	1	1	0.092	271719		
Template	Module						Subarray					
	B						SUB64P					
Spectral Elements	#	Short Pupil	Short Filter	Long Pupil	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F162M	F150W2	CLEAR	F480M	RAPID	5	10	1	10	3.035	

Proposal 10893 - Observation 4 - A High-Precision Test of Substellar Cooling Models with the Benchmark System HR 7672AB

Special Requirements	Aperture PA Range 73 to 83 Degrees (V3 72.6974475 to 82.6974475) Aperture PA Range 103 to 118 Degrees (V3 102.6974475 to 117.6974475) Time Series Observation No Parallel Attachments Group Observations 4, 5, 6, 7, 8, 9, Non-interruptible Aperture PA Offset 4 from 7 by 7 to 14 Degrees (Same offsets in V3)
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Proposal 10893 - Observation 5 - A High-Precision Test of Substellar Cooling Models with the Benchmark System HR 7672AB

Observation	Proposal 10893, Observation 5: HR7672-F140M-Roll1											Fri Mar 13 20:02:15 GMT 2026
	Diagnostic Status: Warning Observing Template: NIRCam Time Series											
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. (HR7672-F140M-Roll1 (Obs 5)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	HR7672	RA: 20 04 6.2208 (301.0259200d) Dec: +17 04 12.68 (17.07019d) Equinox: J2000				Proper Motion RA: -387.472 mas/yr Proper Motion Dec: -419.4970000298781 mas/yr Parallax: 0.0562693" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Star Description=[Brown dwarfs, G qiants]											
Acquisition	#	Target	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID		
	1	SAME	SUB32TATS	F405N+F444W	RAPID	5	1	1	0.092	271719		
Template	Module						Subarray					
	B						SUB64P					
Spectral Elements	#	Short Pupil	Short Filter	Long Pupil	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Integrations	Total Exposure Time	Optional ETC ID
	1	CLEAR	F140M	CLEAR	F480M	RAPID	5	8	1	8	2.428	

Proposal 10893 - Observation 5 - A High-Precision Test of Substellar Cooling Models with the Benchmark System HR 7672AB

Special Requirements

Aperture PA Range 73 to 83 Degrees (V3 72.6974475 to 82.6974475)
Aperture PA Range 103 to 118 Degrees (V3 102.6974475 to 117.6974475)
Time Series Observation
No Parallel Attachments

Group Observations 4, 5, 6, 7, 8, 9, Non-interruptible
Aperture PA Offset 5 from 8 by 7 to 14 Degrees (Same offsets in V3)

Proposal 10893 - Observation 6 - A High-Precision Test of Substellar Cooling Models with the Benchmark System HR 7672AB

Observation	Proposal 10893, Observation 6: HR7672-F115W-Roll1											Fri Mar 13 20:02:15 GMT 2026
	Diagnostic Status: Warning Observing Template: NIRCam Time Series											
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. (HR7672-F115W-Roll1 (Obs 6)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	HR7672	RA: 20 04 6.2208 (301.0259200d) Dec: +17 04 12.68 (17.07019d) Equinox: J2000				Proper Motion RA: -387.472 mas/yr Proper Motion Dec: -419.4970000298781 mas/yr Parallax: 0.0562693" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Star Description=[Brown dwarfs, G qiants]											
Acquisition	#	Target	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID		
	1	SAME	SUB32TATS	F405N+F444W	RAPID	5	1	1	0.092	271719		
Template	Module					Subarray						
	B					SUB64P						
Spectral Elements	#	Short Pupil	Short Filter	Long Pupil	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Integrations	Total Exposure Time	Optional ETC ID
	1	CLEAR	F115W	CLEAR	F480M	RAPID	3	8	1	8	1.626	

Proposal 10893 - Observation 6 - A High-Precision Test of Substellar Cooling Models with the Benchmark System HR 7672AB

Special Requirements	Aperture PA Range 73 to 83 Degrees (V3 72.6974475 to 82.6974475) Aperture PA Range 103 to 118 Degrees (V3 102.6974475 to 117.6974475) Time Series Observation No Parallel Attachments Group Observations 4, 5, 6, 7, 8, 9, Non-interruptible Aperture PA Offset 6 from 9 by 7 to 14 Degrees (Same offsets in V3)
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Proposal 10893 - Observation 7 - A High-Precision Test of Substellar Cooling Models with the Benchmark System HR 7672AB

Observation	Proposal 10893, Observation 7: HR7672-F162M-Roll2											Fri Mar 13 20:02:15 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRCam Time Series											
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(HR7672-F162M-Roll2 (Obs 7)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	HR7672	RA: 20 04 6.2208 (301.0259200d) Dec: +17 04 12.68 (17.07019d) Equinox: J2000				Proper Motion RA: -387.472 mas/yr Proper Motion Dec: -419.4970000298781 mas/yr Parallax: 0.0562693" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Star Description=[Brown dwarfs, G qiants]											
Acquisition	#	Target	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID		
	1	SAME	SUB32TATS	F405N+F444W	RAPID	5	1	1	0.092	271719		
Template	Module					Subarray						
	B					SUB64P						
Spectral Elements	#	Short Pupil	Short Filter	Long Pupil	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F162M	F150W2	CLEAR	F480M	RAPID	5	10	1	10	3.035	

Proposal 10893 - Observation 7 - A High-Precision Test of Substellar Cooling Models with the Benchmark System HR 7672AB

Special Requirements	Time Series Observation No Parallel Attachments Group Observations 4, 5, 6, 7, 8, 9, Non-interruptible Aperture PA Offset 4 from 7 by 7 to 14 Degrees (Same offsets in V3)
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Proposal 10893 - Observation 8 - A High-Precision Test of Substellar Cooling Models with the Benchmark System HR 7672AB

Observation	Proposal 10893, Observation 8: HR7672-F140M-Roll2											Fri Mar 13 20:02:15 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRCam Time Series											
Diagnostics	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(HR7672-F140M-Roll2 (Obs 8)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	HR7672	RA: 20 04 6.2208 (301.0259200d) Dec: +17 04 12.68 (17.07019d) Equinox: J2000				Proper Motion RA: -387.472 mas/yr Proper Motion Dec: -419.4970000298781 mas/yr Parallax: 0.0562693" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Star Description=[Brown dwarfs, G qiants]											
Acquisition	#	Target	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID		
	1	SAME	SUB32TATS	F405N+F444W	RAPID	5	1	1	0.092	271719		
Template	Module					Subarray						
	B					SUB64P						
Spectral Elements	#	Short Pupil	Short Filter	Long Pupil	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Integrations	Total Exposure Time	Optional ETC ID
	1	CLEAR	F140M	CLEAR	F480M	RAPID	5	8	1	8	2.428	

Proposal 10893 - Observation 8 - A High-Precision Test of Substellar Cooling Models with the Benchmark System HR 7672AB

Special Requirements	Time Series Observation No Parallel Attachments Group Observations 4, 5, 6, 7, 8, 9, Non-interruptible Aperture PA Offset 5 from 8 by 7 to 14 Degrees (Same offsets in V3)
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Proposal 10893 - Observation 9 - A High-Precision Test of Substellar Cooling Models with the Benchmark System HR 7672AB

Observation	Proposal 10893, Observation 9: HR7672-F115W-Roll2											Fri Mar 13 20:02:15 GMT 2026
	Diagnostic Status: Warning Observing Template: NIRCam Time Series											
Diagnostics	(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. (HR7672-F115W-Roll2 (Obs 9)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	HR7672	RA: 20 04 6.2208 (301.0259200d) Dec: +17 04 12.68 (17.07019d) Equinox: J2000				Proper Motion RA: -387.472 mas/yr Proper Motion Dec: -419.4970000298781 mas/yr Parallax: 0.0562693" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Star Description=[Brown dwarfs, G qiants]											
Acquisition	#	Target	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID		
	1	SAME	SUB32TATS	F405N+F444W	RAPID	5	1	1	0.092	271719		
Template	Module					Subarray						
	B					SUB64P						
Spectral Elements	#	Short Pupil	Short Filter	Long Pupil	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Integrations	Total Exposure Time	Optional ETC ID
	1	CLEAR	F115W	CLEAR	F480M	RAPID	3	8	1	8	1.626	

Proposal 10893 - Observation 9 - A High-Precision Test of Substellar Cooling Models with the Benchmark System HR 7672AB

Special Requirements	Time Series Observation No Parallel Attachments Group Observations 4, 5, 6, 7, 8, 9, Non-interruptible Aperture PA Offset 6 from 9 by 7 to 14 Degrees (Same offsets in V3)
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